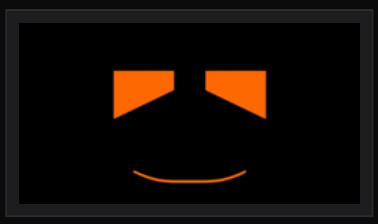
AUTONOMOUS COMPANION ROBOT · RASPBERRY PI · BUILT BY HARSH

FRIDAY

Hi, I'm **Friday**. Harsh built me — the chassis, the wiring and the code. I wake when you say my name, I look before I move, I keep what you teach me, and if I don't know something, I'll just say so.



MY FACE · 7″ SCREEN



FRIDAY'S ACTUAL BODY · 3D-PRINTED



OUT FOR A DRIVE

BODY

- **Raspberry Pi** — the brain box
- **360° LiDAR** — on the roof
- **2 cameras** — room + floor
- **8×8 ToF** — depth at the bumper
- **7" screen** — her face
- **Mic + speaker** — ears and voice
- **4 motors** — L298N drive

20 FACES

She picks the one that fits what she just said — safety faces always win.



01 Brain & Voice

- **"Hey Friday"** wake word, double-checked
- Speech-to-text on the Pi itself
- Natural voice — English, Hindi or Hinglish
- Claude brain with **live web search**
- Knows when the TV is talking, not you
- Drive, stop, go home, explore — **offline**

02 Vision

- Two cameras: the room and the floor
- Object detection, always updating
- Notices something new — and mentions it once
- **"What do you see?"** — she looks first, every time
- Tells you what she saw after exploring

03 Navigation

- Maps the room with a 360° LiDAR
- Finds herself on her saved map
- **"Remember this place" → "go to the kitchen"**
- Go home, come back, try again
- Explores on her own, then **returns to her start**
- Backs off when stuck — and asks for help

04 Assistant & Memory

- Remembers facts between conversations
- **Memory Lane:** what she knows about people
- Alarms, timers, clock & weather on her screen
- Phone dashboard: cameras, maps, logs
- Daily spend cap, so she can't run up a bill

05 Self-Improvement

- Correct her once — she keeps it
- Notes what she didn't know...
- ...and looks it up quietly when idle
- Reflects on her day; **"how much have you grown?"**
- Edits her own code — tested, safety locked

06 Study Companion

- Drop in your PDF notes
- Ask questions — answered from **your** notes
- Flashcards with spaced repetition (FSRS)
- Knows which cards are due, and when
- Study mode on the phone dashboard

SAFETY

Every move is guarded by the server, not trusted to the brain · a bumper depth sensor stops her in centimetres · **"stop" always wins** · no wheel encoders, so she says **"about"** and means it · when truly stuck she stops and waits for a person.

> **Before I move, I look. Every single time.**
> — FRIDAY

> **"I don't know" is an answer. A guess is how robots get broken.**
> — FRIDAY

> **Kuch naya dikha? Chalo, dekh ke aate hain.**
> — FRIDAY · saw something new? let's go look

> *The world is its own best model.*
> — RODNEY BROOKS, 1990

57	20	360°	2	8×8	1
TOOLS	FACES	LIDAR	CAMERAS	DEPTH GRID	RASPBERRY PI

BUILT BY HARSH

Friday · 2026